
Please circle your Class Section: C01 (Dr. Athar) C02 (Dr. Negm)

Name: _____ Student ID: _____

ELECENG 3CL4 (Introduction to Control Systems)
Midterm Examination

Department of Electrical and Computer Engineering, McMaster University
Instructors: Dr. Shahrukh Athar (C01) and Dr. Ayman Negm (C02), Winter 2024
Wednesday, March 6, 2024, 6:30 PM to 8:30 PM (120 minutes)

Examination Conditions This test is conducted under standard examination conditions. In particular:

- You are not to communicate in any manner with any other student.
- You are not to look at the questions nor start writing until told to do so by an invigilator.
- When the invigilator announces that the test is over, you are to stop writing *immediately*.

The penalties for violating these or any other standard examination conditions will be severe, including a zero on the test and the commencement of formal disciplinary proceedings.

Instructions

- There are five (5) questions on this examination paper. Answer all five questions in the space provided in this exam paper. You may use the back of the pages if necessary.
- When given permission to start writing, you must first ensure that your copy of the examination paper is complete and bring any discrepancies to the attention of your exam invigilator.
- Write your name and student number on the front page of this examination paper and on any additional booklet that you include with it.
- Write your answers neatly and in a logical fashion. We may refuse to mark answers which are difficult to decipher.
- Direct answers without proper justifications will lead to reduced grade. However, justifications should be concise and to-the-point.
- You may write your test in pencil, but booklets written in pencil or with pages torn out will not be remarked.
- Closed-books, closed-notes.
- Aids permitted:

One single-sided (8.5 x 11) crib sheet with only handwritten notes (Not printed/photocopied).

McMaster Standard Calculator (Casio FX-991MS/MS+).

- Total Marks: 100 (mark allocations for questions/parts are shown within questions).

Q1 (20)	Q2 (20)	Q3 (20)	Q4 (20)	Q5 (20)	Total (100)

Useful Information

- The solutions to the quadratic equation $ax^2 + bx + c = 0$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- In the function $f(t) = Ae^{-t/\tau}$ the term τ is sometimes referred to as the time constant.
- The following Laplace transform pairs may be of use. Here $u(t)$ is the unit step function

$$\begin{aligned}
 u(t) &\longleftrightarrow \frac{1}{s} \\
 e^{-at}u(t) &\longleftrightarrow \frac{1}{s+a} \\
 \sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{s^2 + \omega^2} \\
 \cos(\omega t)u(t) &\longleftrightarrow \frac{s}{s^2 + \omega^2} \\
 e^{-at} \cos(\omega t)u(t) &\longleftrightarrow \frac{s+a}{(s+a)^2 + \omega^2} \\
 e^{-at} \sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{(s+a)^2 + \omega^2} \\
 t^n u(t) &\longleftrightarrow \frac{n!}{s^{n+1}} \\
 \frac{df(t)}{dt} &\longleftrightarrow sF(s) - f(0^-) \\
 \frac{d^2 f(t)}{dt^2} &\longleftrightarrow s^2 F(s) - sf(0^-) - f'(0^-) \\
 \int_{0^-}^t f(\lambda) d\lambda &\longleftrightarrow \frac{F(s)}{s}
 \end{aligned}$$

- A generic form for a rational Laplace Transform is $G(s) = K_G \frac{\prod_{i=1}^M (s+z_i)}{\prod_{j=1}^n (s+p_j)}$, where the set of finite zeros is $\{-z_i\}_{i=1}^M$ and the set of poles is $\{-p_j\}_{j=1}^n$.
- The final value theorem states that for a Laplace Transform $Y(s)$ with all its poles in the left half plane, except for at most one pole at the origin, $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$.
- For a standard second-order system with a transfer function of the form $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$, when $0 < \zeta < 1$, the poles are at positions $-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$. The step response of such a system has the following properties:

- $y_{\text{step}}(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin\left(\omega_n\sqrt{1-\zeta^2}t + \theta\right)$, where $\theta = \cos^{-1}(\zeta)$.
- 2% settling time, $T_s \approx \frac{4}{\zeta\omega_n}$.
- Percentage (maximum) overshoot, $P.O. = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}$. Hence, $\zeta = \frac{-\ln(P.O./100)}{\sqrt{\pi^2 + \ln^2(P.O./100)}}$.
- Time of peak, $T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$.
- For $0.3 \leq \zeta \leq 0.8$, the 10%–90% rise time is $T_r \approx \frac{2.16\zeta + 0.6}{\omega_n}$.

- Steady state error constants:

- Position error constant: $K_{\text{posn}} = \lim_{s \rightarrow 0} G_c(s)G(s)$
- Velocity error constant: $K_v = \lim_{s \rightarrow 0} sG_c(s)G(s)$
- Acceleration error constant: $K_a = \lim_{s \rightarrow 0} s^2 G_c(s)G(s)$

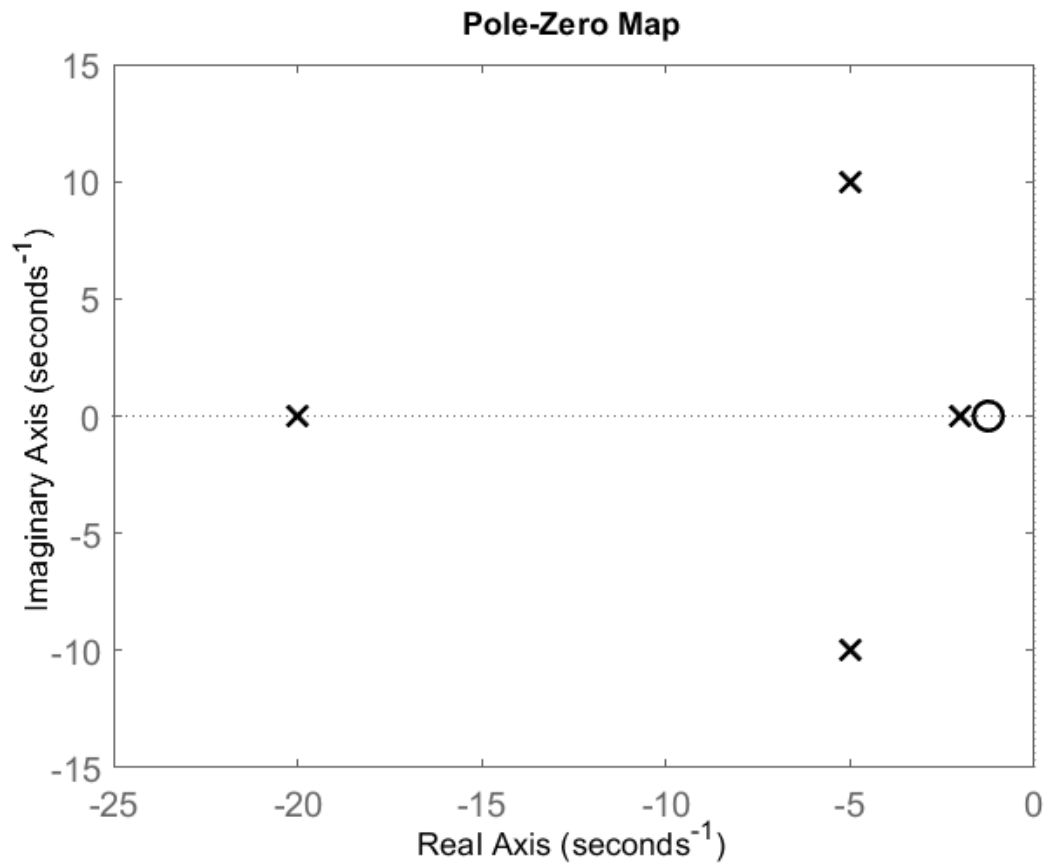
Question 1 (20 marks). This question is composed of five short parts (a-e). **Answers are expected to be short and concise.** We have provided more space than is necessary. Answer the following questions:

a) (4 marks) Given the following Routh-Hurwitz table:

Row 5	1	-3	-5
Row 4	2	4	6
Row 3	-5	-8	
Row 2	0.8	6	
Row 1	29.5		
Row 0	6		

Determine the number of closed-loop poles in the left half plane, in the right half plane and on the $j\omega$ -axis.

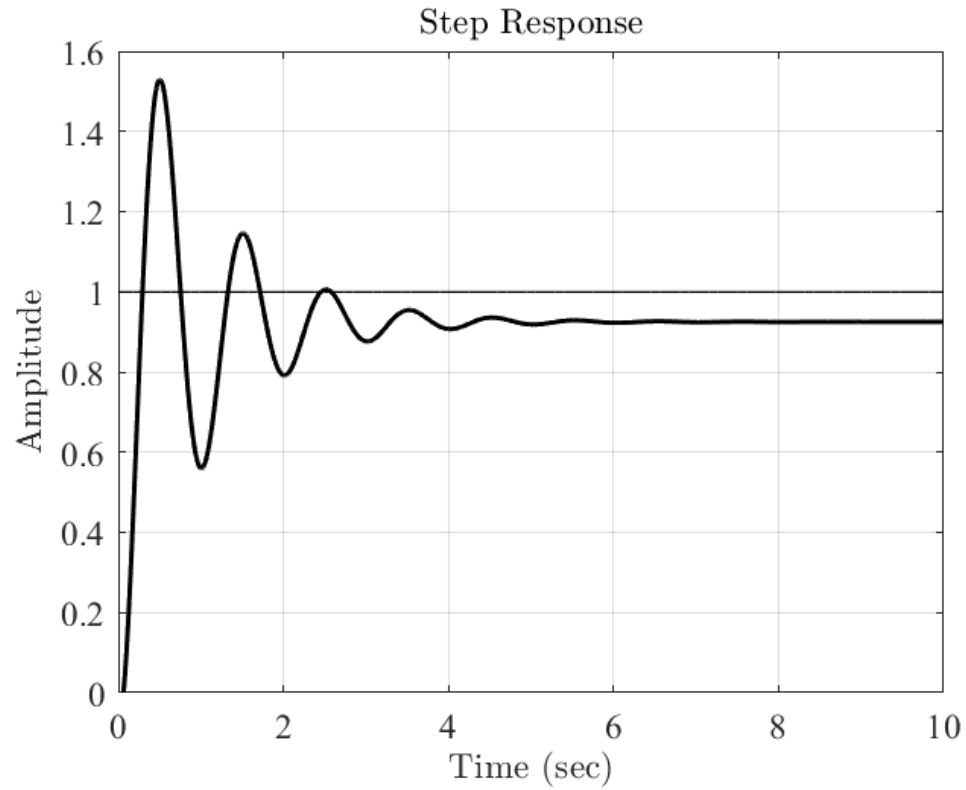
- b) (4 marks) The locations of the poles and zero of a fourth order closed-loop system are shown in the figure below. You are required to approximate the behavior of this system with either a first-order or a second-order system. Which one will you choose? Justify your answer.



c) (4 marks) Given a system with an open-loop transfer function of the form:

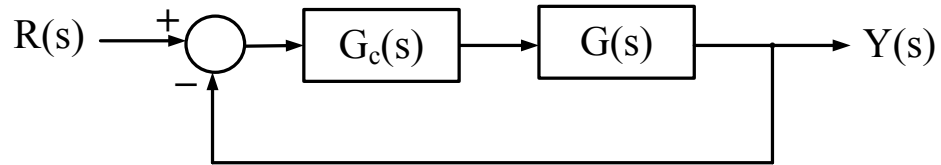
$$L(s) = \frac{K(s+a)}{s(s+b)(s+c)}$$

where $a > 0$, $b > 0$, and $c > 0$. Can such a system have a closed-loop step response as depicted in the figure below? Assume that steady-state is achieved around $t = 6$ sec. Justify your answer.



d) (4 marks) Consider the following feedback control system in which:

$$G(s) = \frac{4}{(s-2)(s+1)}, \quad G_c(s) = \frac{k(s-2)}{s+4}$$

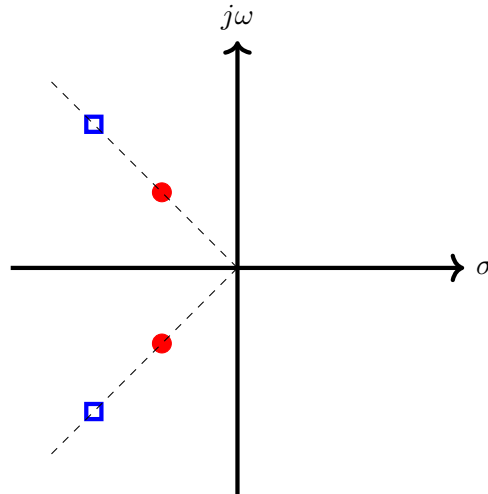


Explain whether it is practical to use this controller to stabilize the system.

e) (4 marks) A standard second-order system has a transfer function of the form:

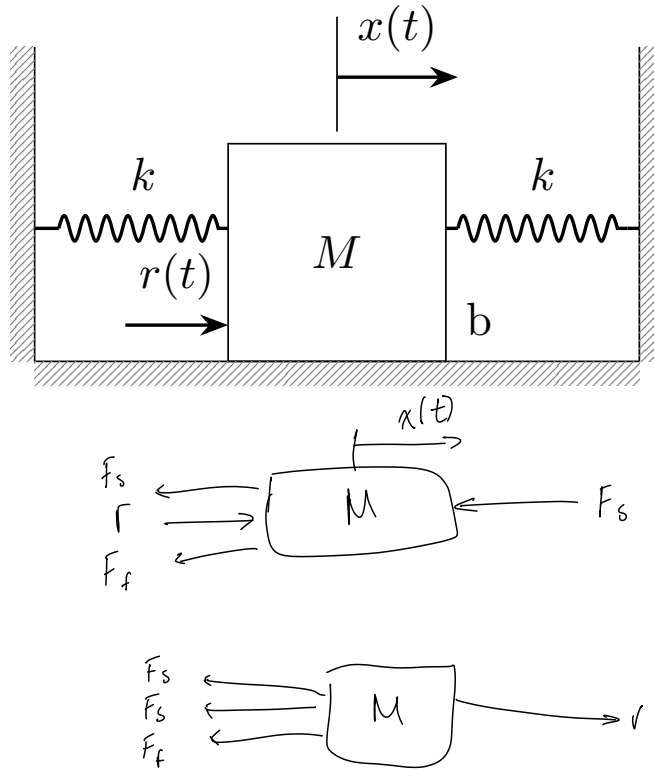
$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

In the figure below, the locations of two pairs of poles from two such systems are displayed in the s -plane. This question is concerned with the step responses of these systems. Rank the two systems based on the values (from the largest to smallest) of each of the following properties of the step response: (1) percentage (maximum) overshoot; (ii) settling time; and (iii) peak time. Note that no numerical value is provided so your comparisons should be in relative terms.



Question 2 (20 marks). The figure shows a mass ($M = 40 \text{ kg}$) that is attached to walls by two linear springs of spring constant k , where $k = 20 \text{ N/m}$. The mass is placed on a rough surface of viscous friction coefficient b . An external force $r(t)$ is acting upon the mass. The displacement $x(t)$ is measured from the equilibrium point where both springs are relaxed before applying the external force.

- (6 marks) Derive a differential equation that describes the motion of the mass.
- (5 marks) Find the transfer function $\frac{X(s)}{R(s)}$.
- (5 marks) Find the steady-state values of $x(t)$ and $\frac{dx(t)}{dt}$ when a constant external force of 1 N is applied.
- (4 marks) If the external force is removed, find the minimum value of b that will prevent the mass from oscillating.



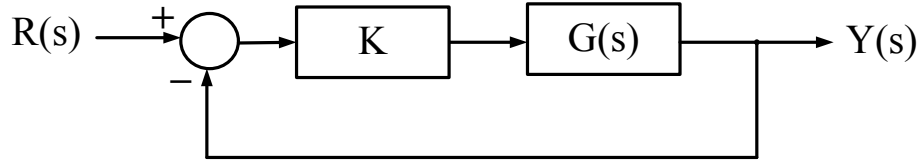
$$F(t) = ma(t) = r(t) - 2 \cdot F_s - F_f$$

[Additional space for Question 2.]

Question 3 (20 marks). Consider a second-order system with the generic transfer function, $G(s)$, given below:

$$G(s) = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$

This (open-loop) system is placed in a unity gain negative feedback loop (i.e., the loop in the figure below with $K = 1$).

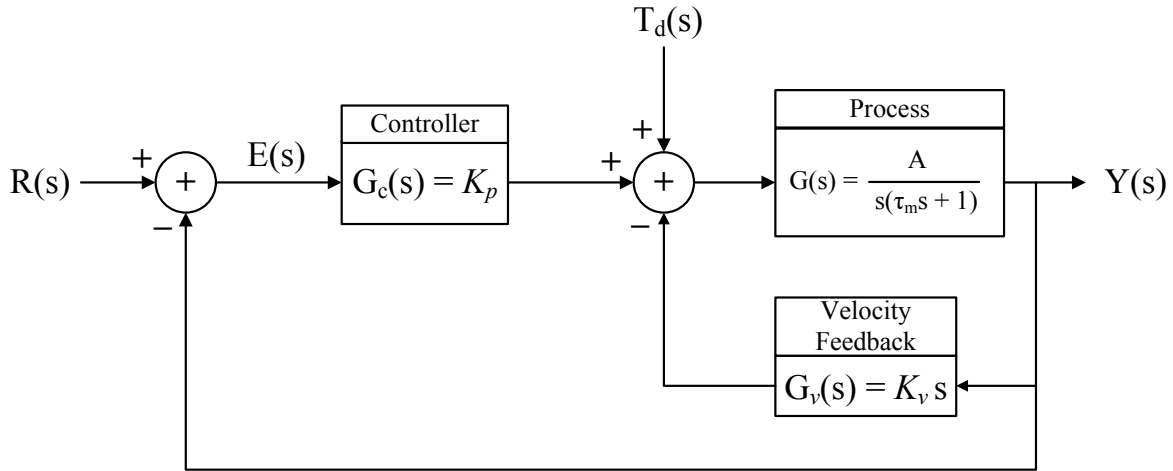


The step response of the closed-loop system has a percentage overshoot that is exactly 30% and a 2% settling time that is equal to 0.25 seconds.

- (6 marks)** Find the angle θ that the closed-loop poles make with the negative real axis in the s -plane.
- (6 marks)** Find the location of the closed-loop poles in the s -plane. Apart from stating the location of the poles, you should also provide a sketch that depicts them in the s -plane.
- (4 marks)** Find the peak time of this system.
- (4 marks)** If K is doubled, find the new value of the step response percentage overshoot.

[Additional space for Question 3.]

Question 4 (20 marks).



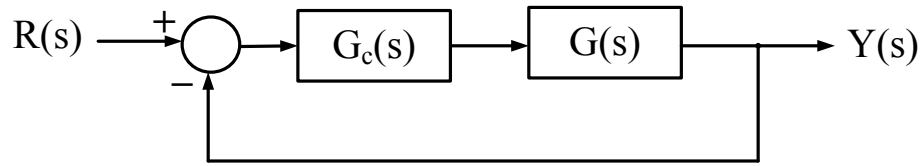
Consider the closed loop feedback control system above. The total output response to the reference and disturbance inputs is given by:

$$Y(s) = \frac{\frac{K_p A}{\tau_m}}{s^2 + \frac{1+K_v A}{\tau_m} s + \frac{K_p A}{\tau_m}} R(s) + \frac{\frac{A}{\tau_m}}{s^2 + \frac{1+K_v A}{\tau_m} s + \frac{K_p A}{\tau_m}} T_d(s)$$

- (5 marks)** With $T_d(s) = 0$, derive expressions for the damping ratio ζ and natural frequency ω_n .
- (5 marks)** How do controller parameters K_p and K_v affect settling time and percentage overshoot.
- (7 marks)** Find an expression for the magnitude of the steady-state error, e_{SS} , due to a step input $R(s) = \frac{\theta_d}{s}$ and a step disturbance $T_d(s) = \frac{\tau_d}{s}$.
- (3 marks)** How do the controller parameters K_p and K_v affect the steady-state error due to a step disturbance?

[Additional space for Question 4.]

Question 5 (20 marks).



Consider the closed-loop feedback control system above where $G_c(s) = \frac{K}{s+5}$ and $G(s) = \frac{1}{s(s+3)(s+4)}$.

- a) **(4 marks)** Derive the transfer function of the closed-loop system, $\frac{Y(s)}{R(s)}$.
- b) **(8 marks)** Identify the characteristic equation and construct the complete Routh-Hurwitz table for this system.
- c) **(3 marks)** Find the range of K for which the system is stable.
- d) **(5 marks)** What value of K puts a pair of closed-loop poles on the $j\omega$ -axis, what is the position of the $j\omega$ -axis poles?

[Additional space for Question 5.]